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To solve using a proportion, be sure that the two ratios have corresponding units.

At Bright's Movie Theater, 3 tickets cost \$24. How much would 4 tickets cost?

Write a ratio with two numbers from the problem. Include labels.

$$\begin{array}{r} 3 \text{ tickets} \\ \hline \$24 \end{array}$$

Write a proportion, using a variable for the number you do not know.

$$\frac{3 \text{ tickets}}{\$24} = \frac{4 \text{ tickets}}{\$p}$$

Multiply to get the cross products and find the unknown value.

$$\begin{aligned} 3 \times p &= 24 \times 4 \\ 3 \times p &= 96 \\ p &= 96 \div 3 \\ p &= \text{\$}32 \end{aligned}$$

The cost of 4 tickets is \$32.

Another way to set up a proportion for this problem is this: $\frac{\$24}{3 \text{ tickets}} = \frac{\$p}{4 \text{ tickets}}$

It doesn't matter which number goes on top or bottom in the first ratio. What *does* matter is that both ratios are set up in the same order with corresponding units.

For example, in the problem above, writing the two ratios as

$$\frac{\$24}{3 \text{ tickets}} = \frac{4 \text{ tickets}}{\$p}$$

would *not* be a true proportion because they do not compare the same units in their numerators or the same units in their denominators.

Another way to express the same proportion is

$$\frac{\$24}{\$p} = \frac{3 \text{ tickets}}{4 \text{ tickets}}$$

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Lesson 6: Ratios and Proportions

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